

LESSON 1.4: Mailbox Management

What this lesson is about

This lesson introduces you to the usage of Sterling B2B Integrator mailbox in Sterling File Gateway. It also discusses the communication of information through file transfers between a producer and a consumer.

What you should be able to do

- Describe Sterling B2B Integrator mailboxes.
- Identify the location of the producer and consumer mailbox.
- Extract data from a mailbox of producer and consumer.
- Explain the importance of global mailbox in Sterling File Gateway.

How you will check your progress

The progress of the course is analyzed based on the successful application of the hands-on exercises.

Additional References

Documentation

- Mailboxes:
http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_overview.doc/SFGMailboxes.html?lang=en
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Sterling B2B Integrator mailboxes

Introduction

The Sterling B2B Integrator Mailbox system is a secure business document repository with a store-and-forward communication infrastructure. The Sterling B2B Integrator Mailbox adds to the business automation and facilities of Sterling B2B Integrator to conveniently build B2B electronic commerce communities.

In addition, the Sterling B2B Integrator Mailbox provides a trading partner browser interface that is called the Mailbox Browser Interface (MBI). No additional software is required when the Sterling B2B Integrator Mailbox is deployed to an electronic trading community. The MBI is secure and simple to use, requiring no special user training.

Sterling B2B Integrator Mailbox Components

The Sterling B2B Integrator Mailbox system that is managed by Sterling File Gateway includes the following components:

- Mailbox Browser Interface (MBI)
- Messages
- Mailbox
- Archive

Mailbox Browser Interface (MBI)

The MBI allows Internet users to securely access their mailboxes from a standard web browser. The MBI is a web application and therefore requires only a browser.

Users can send messages to, retrieve messages from, and search messages in any mailbox for which they have permissions. The MBI allows users to change their account password.



Note

Sterling File Gateway uses the mailbox system that is present in Sterling B2B Integrator. This lesson is provided in order to give students an overview of the mailboxing system in Sterling B2B Integrator.

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Sterling B2B Integrator mailboxes

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Messages

Business documents that are stored in the Sterling B2B Integrator Mailbox are called *messages*. Messages can contain business content in any format such as binary, EDI, or XML.

Each message is assigned an extraction policy that specifies the rules for extracting messages from a mailbox.

A message added to a mailbox can be processed immediately or at a scheduled time in the future.

Mailbox

A *mailbox* is a storage area for business documents and provides an administrative hierarchy that is easy to manage and understand.

Mailboxes form a hierarchy. The top of the hierarchy is called the root mailbox. A slash (/) denotes the root mailbox. Mailboxes are able to support sub-mailboxes. This organizational concept is known as mailbox hierarchy.

System users have access to their documents while administrators can organize and manage documents across all mailboxes.

Managing Mailbox Messages

Introduction

A key function of Sterling File Gateway is the routing of files from one mailbox to another. Sterling File Gateway uses the mailbox system within the Sterling B2B Integrator application for its routing of messages.

The Sterling B2B Integrator Mailbox system manages messages using routing rules and services. Routing rules are used to manage business document processing.

Routing rules are used to schedule processing the business document and can be configured for automatic, scheduled, or manual processing of the message. Services are used to add messages to a particular mailbox, locate messages, delete messages, extract messages, and schedule the routing rule evaluation.

Mailbox Creation

Mailboxes that are used for Sterling File Gateway operations can be created automatically in the following circumstances:

- When a Partner is created in Sterling File Gateway.
- When a Routing Channel is created.
- When Partners are migrated from Advanced File Transfer.
- When Partners are imported into Sterling B2B Integrator.
- At file transfer runtime (by dynamic routing channel templates).

When a Partner is created in Sterling File Gateway

When a Partner is created, a mailbox with the same name is created. For example, if the partner is named *PartnerOne* a mailbox is created where the mailbox absolute path is */PartnerOne*. When the *PartnerOne* user logs in, the absolute path is not displayed. The *PartnerOne* user sees the virtual path only, which is in this example *"/*.

When a Routing Channel is created

When a routing channel is created, a producer mailbox is created in the structure specified in the governing Routing Channel Template. Creation of a Routing Channel need not create a consumer mailbox.

A consumer mailbox can be created on demand (at runtime) if configured to do so in the Routing Channel Template. Whether created at Routing Channel creation time or on demand at run time, the consumer mailbox conforms to the structure mandated by the Routing Channel Template.

When creating Routing Channels, Sterling File Gateway checks to determine whether a mailbox of the required structure exists. The Sterling File Gateway creates the mailbox along with any required permissions if it does not exist.

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Managing Mailbox Messages

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Routing rules

Routing rules provide a powerful and easy-to-use mechanism for controlling document routing. Additionally, the Sterling B2B Integrator Mailbox is easily combined with native Sterling B2B Integrator capabilities such as business processes, document translation, B2B protocols, and backend system integration.

You can create routing rules to initiate action automatically when a message is added to a mailbox. Automatic routing assures that the rule is evaluated at least once. When a message is added to a mailbox, any rules established for the Sterling B2B Integrator Mailbox are evaluated automatically. Later, any matching rules are executed. In some situations, as a result of an outage, a message can be routed once before the outage and again at system restart.

Manual and scheduled evaluation do not provide a guarantee that message is delivered at least once. Routing usually occurs once but if a system outage occurs, the routing does not resume at system start.

Routing rule and pattern and action

Routing rules consist of two parts:

- A pattern to specify when the rule applies.
- An action the rule must perform.

Routing rules contain string patterns that are used to match message names. This string can contain wild card characters. You can restrict the application of the rule to specific mailboxes.

When a message name matches the string pattern of a rule, the rule triggers the execution of a designated Sterling B2B Integrator business process. That process can perform any processing, including extraction of the message and integration with backend systems or a notification action.



Example

A routing rule can trigger a business process that pages an administrator whenever a message arrives in the Dead Letter mailbox.

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Managing Mailbox Messages

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Routing rule evaluation mode

A routing rule evaluation mode is used to define when the rule is evaluated. You establish the evaluation mode when you define the routing rule. The following list describes the rule evaluation modes:

- **Manual:** The rule is evaluated when an administrator requests evaluation (Test mode). The same batch of message ID is pulled each time the rule is started.
- **Scheduled:** A schedule is established to evaluate rules. The message ID is pulled once upon success request.

When a routing rule is evaluated, the Sterling B2B Integrator Mailbox searches the mailboxes defined in the rule for message names that match the pattern defined in the rule. When a match is found, the routing rule notifies the business process or contract specified in the routing rule.

Depending on how the business process is defined, the mailbox can extract and process the message, extract the message, and pass it to another application for processing, or notify another application to extract and process the message.

If routing rules are set to be evaluated manually, a user or a business process can evaluate it.

Exercise 1.4.1: Locate mailboxes

Introduction

In this exercise, you will access the Sterling B2B Integrator console to locate the mailboxes that are created as a result of the route channel creation.

Instructions

Complete the following steps to view the mailbox:

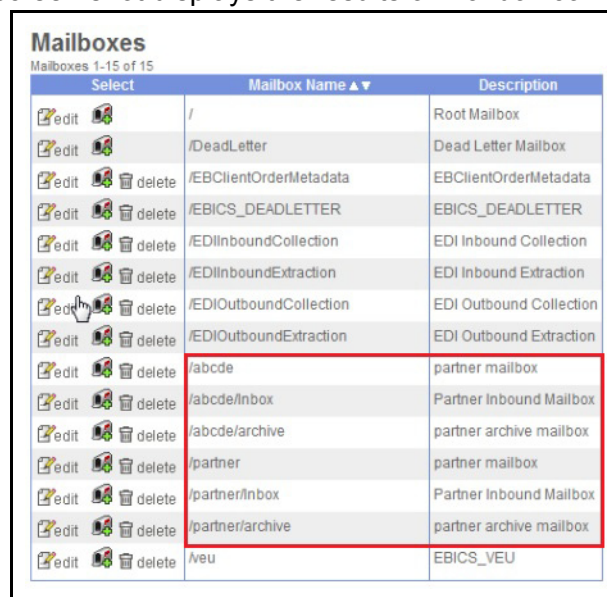
Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: From the **Tools > B2B Console**.

Step 3: From **Administration Menu**, navigate to **Deployment > Mailboxes > Configuration**.

Step 4: From the **Mailbox Configuration** menu, use the **Go!** button under the **List** option to list all mailbox configurations.

Step 5: Locate the mailboxes for **Partner** and **abcde**. The screen shot displays the results of mailbox configuration. The screen shot displays the results of mailbox configuration.



Select	Mailbox Name ▲▼	Description
 	/	Root Mailbox
 	/DeadLetter	Dead Letter Mailbox
 	/EBClientOrderMetadata	EBClientOrderMetadata
 	/EBICS_DEADLETTER	EBICS_DEADLETTER
 	/EDIInboundCollection	EDI Inbound Collection
 	/EDIInboundExtraction	EDI Inbound Extraction
 	/EDIOutboundCollection	EDI Outbound Collection
 	/EDIOutboundExtraction	EDI Outbound Extraction
 	/abcde	partner mailbox
 	/abcde/inbox	Partner Inbound Mailbox
 	/abcde/archive	partner archive mailbox
 	/partner	partner mailbox
 	/partner/inbox	Partner Inbound Mailbox
 	/partner/archive	partner archive mailbox
 	/neu	EBICS_VEU



Note

Make sure that pop-up blocker in your browser is disabled.

Exercise 1.4.2: Enable routing rule schedule

Introduction

In this exercise, you will ensure that an appropriate mailbox routing schedule is enabled. There are two built in routing rules in Sterling B2B Integrator:

- **MailboxEvaluateAllAutomaticRules** - runs every minute and can be edited for longer intervals.
- **MailboxEvaluateAllAutomaticRulesSubMin** - checks for the presence of routable messages every 10 seconds and can be edited for other intervals of less than 1 minute by modifying the MailboxEvaluateAllAutomaticRulesSubMin business process.

Instructions

Complete the steps ensure routing schedule is enabled.

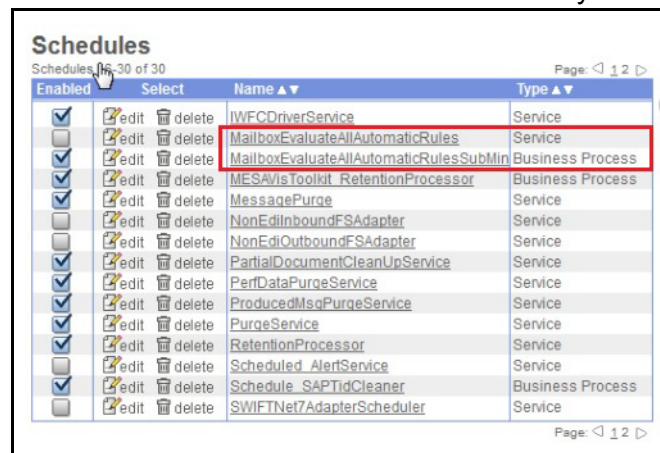
Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: From the **Tools > B2B Console**.

Step 3: From **Administration Menu**, Navigate to **Deployment > Schedules**.

Step 4: Click **Go!** under the **List** option to list all scheduled business process.

Step 5: Locate either **MailboxEvaluateAllAutomaticRules** or **MailboxEvaluateAllAutomaticRulesSubMin** and ensure they are enabled.



Enabled	Select	Name	Type
☒		IWFCDriverService	Service
☒		MailboxEvaluateAllAutomaticRules	Service
☒		MailboxEvaluateAllAutomaticRulesSubMin	Business Process
☒		MES/VisToolkit_RetentionProcessor	Business Process
☒		MessagePurge	Service
☒		NonEdiInboundFSAdapter	Service
☒		NonEdiOutboundFSAdapter	Service
☒		PartialDocumentCleanUpService	Service
☒		PerfDataPurgeService	Service
☒		ProducedMsgPurgeService	Service
☒		PurgeService	Service
☒		RetentionProcessor	Service
☒		Scheduled_AlertService	Service
☒		Schedule_SAPTidCleaner	Business Process
☒		SWIFTNet7AdapterScheduler	Service



Note

You can also view the highlighted service and business process in detail from Service Configuration editor and business process through business process editor.

Additional mailbox information

Disable duplicate messages

An important consideration for mailboxes is related to system behavior regarding duplicate files. A global system setting allows or disallows duplicates (files of the same name) in mailboxes. For optimum system behavior, disallow duplicates in your **customer_overrides.properties** file. This setting ensures that every message in a single mailbox has a unique name. It also ensures that a message and a mailbox has unique name.

If a message is written to a mailbox and the message name matches the name of an existing message in the mailbox, the service deletes the old message before adding the new message.

To edit the `customer_overrides.properties` file:

Step 1: In the **install_dir/properties** directory, locate (or create if necessary) the **customer_overrides.properties** file.

Step 2: Open the **customer_overrides.properties** file in a text editor.

Step 3: Add the following parameter: *mailbox.disallowDuplicateMessages=true*.

Exercise 1.4.3: View disabled duplicate message

Introduction

In this exercise, you will ensure duplicate messages are disallowed in the mailbox properties file. By default, the duplication messages are allowed. If you want to disallow message, you must set disallow duplicate messages parameter name as true by editing or creating the customer_overrides.properties file.

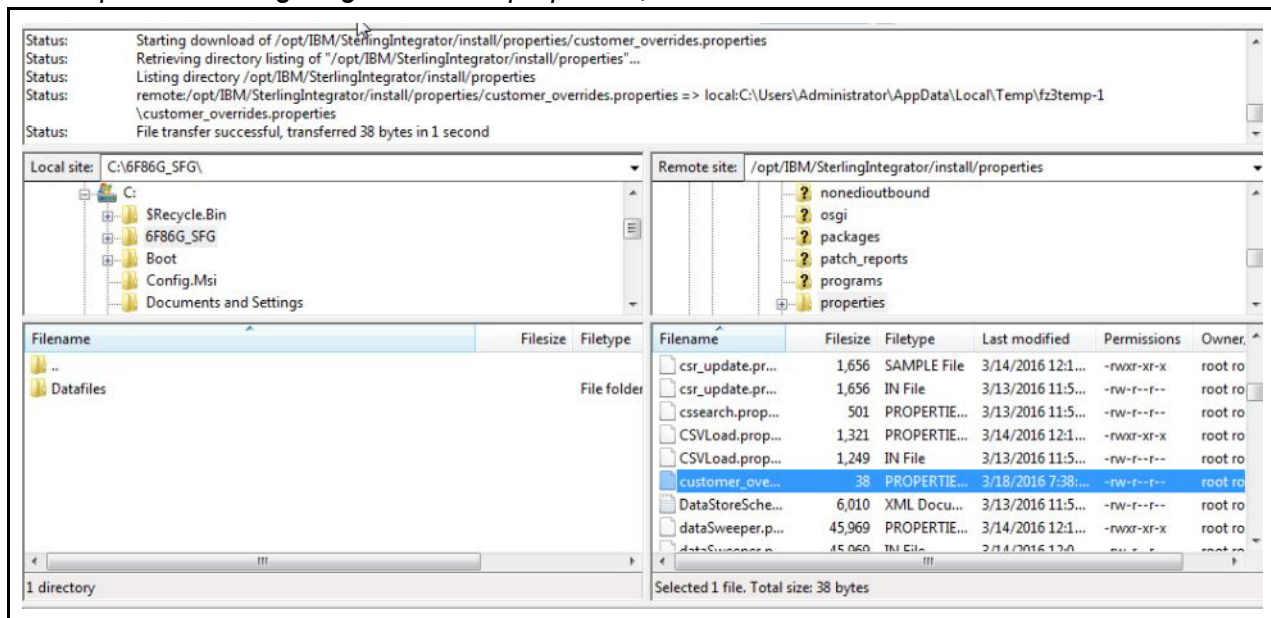
Instructions

Complete the following steps to view the customer_overrides.properties file:

Step 1: Access desktop from windows image, open FileZilla.

Step 2: Select sftp://root@192.168.40.100 from the server drop down list and click **Quickconnect**. You are now connected to the server image.

Step 3: Navigate to the remote panel to Sterling B2B Integrator properties folder */opt/IBM/SterlingIntegrator/install/properties*, as shown in the screen shot.



Step 4: Open the customer_overrides.properties file in a text editor.



Note

If customer_overrides.properties does not exist in the Properties folder of Sterling B2B Integrator, you must create the file and add *mailbox.disallowDuplicateMessages* parameter with a value as True.

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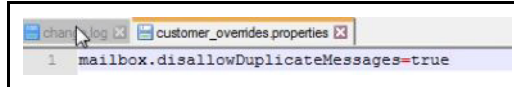
Exercise 1.4.3:View disabled duplicate message

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Instructions

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Step 5: Ensure that the *mailbox.disallowDuplicateMessages* parameter is set to true.



Step 6: Close customer_overrides.properties file.

You must restart Sterling B2B Integrator after editing the properties file. Complete the following steps to restart Sterling B2B Integrator for Linux system:

1. From the <si_install_dir>/bin directory, run the **./hardstop.sh** command to stop Sterling B2B Integrator, click **Enter**.
2. From the <si_install_dir>/bin directory, run the **./setupfiles.sh** command to update the changes of the properties file.
3. From the <si_install_dir>/bin directory, run the **./run.sh** command to start Sterling B2B Integrator.



Note

Ensure that you edit the parameters of the property files in the customer_overrides.properties file. In this exercise, the property disallowDuplicateMessage is set to false in the mailbox.properties file but the same parameter, that is, mailbox.disallowDuplicateMessage is changed to true in the customer_overrides.properties file.

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Additional mailbox information

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Purge messages from mailbox

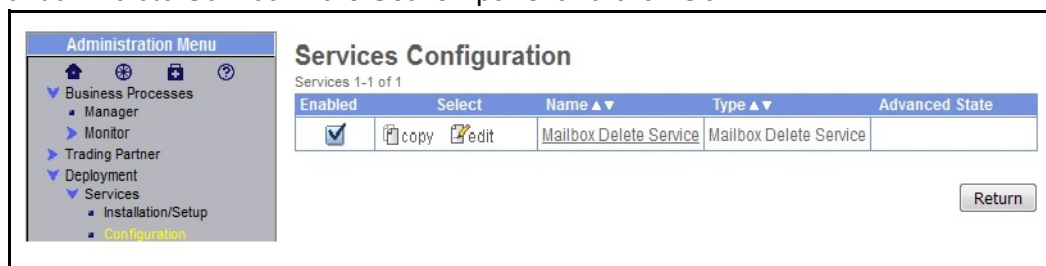
Messages in consumer mailboxes are not automatically purged and over time, they can negatively affect system performance.

To avoid this issue, configure the Mailbox Scheduled Delete service to delete messages from one, many, or all mailboxes. For more information about the Mailbox Scheduled Delete service refer to the following section in the IBM Knowledge Center:

http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.svcs_adpts_m_z.doc/Mailbox_Scheduled_Delete_svc.html?lang=en

Complete the following steps to view Scheduled Mailbox Delete Service:

1. Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.
2. Click **Tools > B2B Console**.
3. Navigate to **Business Processes > Deployment > Configuration**.
4. Type *Mailbox Delete Service* in the **Search** panel and click **Go**.



5. Click **edit** icon to view the mailbox delete configuration.



Note

Schedule *Mailbox Delete Service* or *MessagePurge* business process to purge, as per the business requirement.

Distributed mailbox

Introduction

IBM Global High Availability Mailbox is an add-on component of IBM Sterling B2B Integrator and is robust and reliable to store and replicate message data from partners across any geography.

Current approaches to provide high available B2B and MFT environment:

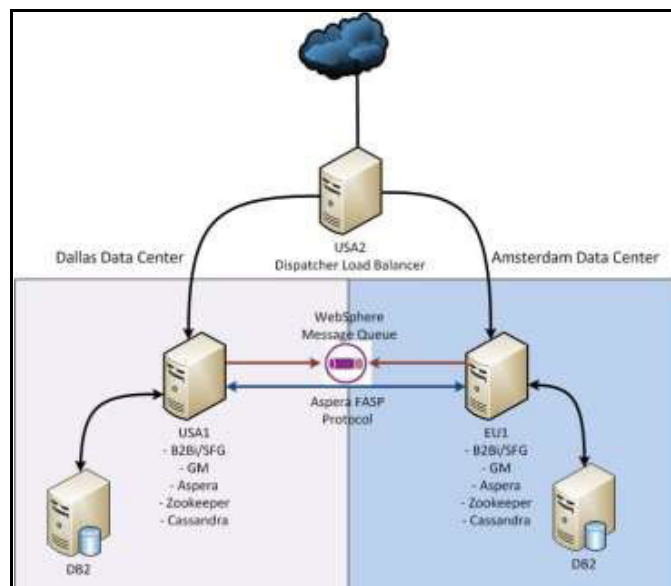
- Shares payload across data centers at the communications layer for various protocols.
- Asynchronous or Synchronous data replication is configurable by mailbox.
- A consistent view is provided across data centers of what files get processed and where regardless of data center status.
- Transaction processing is balanced across distribution centers.
- Integration with Control Center v6.1 to monitor system health.

IBM Global Mailbox has an architecture that utilizes multiple components to provide a synchronization of payloads and processing events across multiple data centers (the current release supports only two data centers). The components consist of IBM and non-IBM solutions. Depending on the deployment, some are optional requirements.

The IBM Global Mailbox deployment shown in the screen shot is a scaled down topology for ease of learning. It utilizes a set of three servers located in two geographically dispersed data centers such as Dallas, TX USA and Amsterdam, Belgium.

For more details on uploading a file to the Sterling mailbox using myfilegateway, refer to the following link:

http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.my_file_gateway.doc/SFGSendFrommySFG.html?lang=en



Lesson review

What you have been able to do

- Describe Sterling B2B Integrator mailboxes.
 - Identify the location of the producer and consumer mailbox.
 - Extract data from a mailbox of producer and consumer.
 - Explain the importance of global mailbox in Sterling File Gateway.
-